

**SGGS Institute of Engineering and Technology, Vishnupuri, Nanded****Academic Year: 2024-25****Semester: II****Examination: Summer Term****Class: Third Year****Branch: Mechanical Engineering****Date: 16/06/2025****Time: 10:00 - 13:00****Subject: Refrigeration and Air Conditioning****Code: PCC-ME309****Maximum Marks: 100****General Instructions:**

- I. This paper contains $2 + 2 = 4$ pages.
- II. Do not attach any chart to the answer sheet.
- III. Figures to right indicate full marks.
- IV. Draw neat & compact sketches wherever necessary.
- V. Assume data wherever required and state them clearly.
- VI. Symbols have their usual meanings.

Q. N.	Question	Marks	CO	BT
1.	As summer is approaching, department is planning to increase the cooling capacity of water cooler fitted at ground floor of our department. Based on the its expected usage, it is required to produce a refrigerating effect of 15 kW to maintain the cold water temperature at $20^{\circ}C$ by rejecting heat to the ambient air at $37^{\circ}C$. Considering the water cooler is operating as reversed Carnot cycle, help the department identifying the power required to run the system.	10	CO5	BT5
2.	Compare throttling versus turbine expansion on the same $p - v$ diagram. Apply steady-flow energy equation to both of them. For each process, explain CoP of refrigeration cycle considering net work.	10	CO1	BT2
3.	Explain the process of due formation in the winter days considering an initial state as super heated. For same binary mixture, which is initially at super heated state, compare the effect on partial vapour pressure in the process of due formation and obtaining $\phi = 100\%$. Explain with the help of $T - S$ diagram.	10	CO5	BT2
4.	In the trial run of newly ordered refrigeration system, operating between pressure limits of 1.7 bar and 9 bar, following observations for the temperatures were made. The refrigerant R134a (density $\rho = 1206 \text{ kg/m}^3$) is responsible for the extraction of heat from water in parallel flow heat exchanger. Considering volumetric flow rate of water as $27 \times 10^{-6} \text{ m}^3/\text{s}$ and refrigerant as 0.8 LPH. The actual power consumption of compressor is 0.45 kW. calculate theoretical and actual COP. - Compressor exit = $67^{\circ}C$ - Evaporator exit = $33^{\circ}C$ - Condenser exit = $32^{\circ}C$ - Water inlet = $41^{\circ}C$ - Expansion exit = $12^{\circ}C$ - Water exit = $33^{\circ}C$	10	CO2	BT4
5.	Consider that, you are using an air conditioner in rainy season during which humidity is above the comfort level. Considering this, it is required to remove 0.004 kg of water vapour per kg of atmospheric air. If the temperature of air after removal of water vapour became $20^{\circ}C$, at this point, determine relative humidity and dew point temperature, assuming that atmospheric conditions are $30^{\circ}C$, $\phi = 55\%$, and pressure is 1.01325 bar.	10	CO5	BT4

Q. N.	Question	Marks	CO	BT															
6.	It is required to make a saturated air at 5°C by passing the mixture through cooling coil. The mixture is initially at 32°C DBT and 20°C WBT, which absorbs 14 kW of energy from the cooling coil when the flow rate is $42.5\text{ m}^3/\text{min}$. Determine DBT and WBT of mixture when leaving the coil, and the coil bypass factor. <i>Indicate the process on Psychrometric chart on your answer sheet and label it properly</i>	10	CO5	BT1															
7.	In an absorption type refrigerator, the heat is supplied to NH_3 generator by condensing steam at 2 bar and 90% dry. The temperature in the refrigerator is to be maintained at -5°C . Find maximum CoP possible. At ambient temperature of 30°C , if the refrigeration load is 20 tonnes and actual CoP is 70% of the maximum CoP, find the mass of steam required per hour.	10	CO5	BT4															
8.	The manufacturer of an air conditioner claims an isentropic efficiency of 75% for one of its units. This unit operates on the normal vapour compression refrigeration cycle and uses R-22 as the working fluid. This efficiency is for the operating conditions when the evaporator saturation temperature is -5°C and the condenser saturation temperature is 45°C . Selected data for R-22 are provided in the following table. Take C_p of R-22 as 1.2 kJ/kgK . <table border="1"> <thead> <tr> <th>$T(^{\circ}\text{C})$</th><th>$P_{\text{sat}}(\text{kPa})$</th><th>$h_f(\text{kJ/kg})$</th><th>$h_g(\text{kJ/kg})$</th><th>$S_g(\text{kJ/kgK})$</th></tr> </thead> <tbody> <tr> <td>-5</td><td>421.2</td><td>38.76</td><td>248.1</td><td>0.9344</td></tr> <tr> <td>45</td><td>1728</td><td>101</td><td>261.9</td><td>0.8682</td></tr> </tbody> </table> a. Determine the heat absorbed by the refrigerant in the evaporator per unit mass of refrigerant. b. Determine the work input to the compressor and the heat rejected in the condenser per unit mass of refrigerant.	$T(^{\circ}\text{C})$	$P_{\text{sat}}(\text{kPa})$	$h_f(\text{kJ/kg})$	$h_g(\text{kJ/kg})$	$S_g(\text{kJ/kgK})$	-5	421.2	38.76	248.1	0.9344	45	1728	101	261.9	0.8682	15	CO2	BT5
$T(^{\circ}\text{C})$	$P_{\text{sat}}(\text{kPa})$	$h_f(\text{kJ/kg})$	$h_g(\text{kJ/kg})$	$S_g(\text{kJ/kgK})$															
-5	421.2	38.76	248.1	0.9344															
45	1728	101	261.9	0.8682															
9.	In an air conditioning plant, an air handling unit supplies a total of $4000\text{ m}^3/\text{min}$ of dry air which comprises by mass 20% of fresh air at 39°C DBT and 26°C WBT and 80% recirculated air at 24°C DBT and $\phi = 50\%$. The air leaves the cooling coil at 12°C saturated condition. Calculate i) Total cooling load and ii) Room heat gain.	15	CO5	BT5															

**** ALL THE BEST ****

CO1: Illustrate the fundamental principles and applications of refrigeration and air conditioning systems.
CO2: Obtain cooling capacity and coefficient of performance by conducting test on vapor compression refrigeration systems.

CO3: Present the properties, applications, and environmental issues of different refrigerants.

CO4: Calculate cooling load for air conditioning systems used for various applications.

CO5: Operate and analyze the refrigeration and air conditioning systems.

BT1: Remember, **BT2:** Understand, **BT3:** Apply, **BT4:** Analyze, **BT5:** Evaluate, **BT6:** Create.

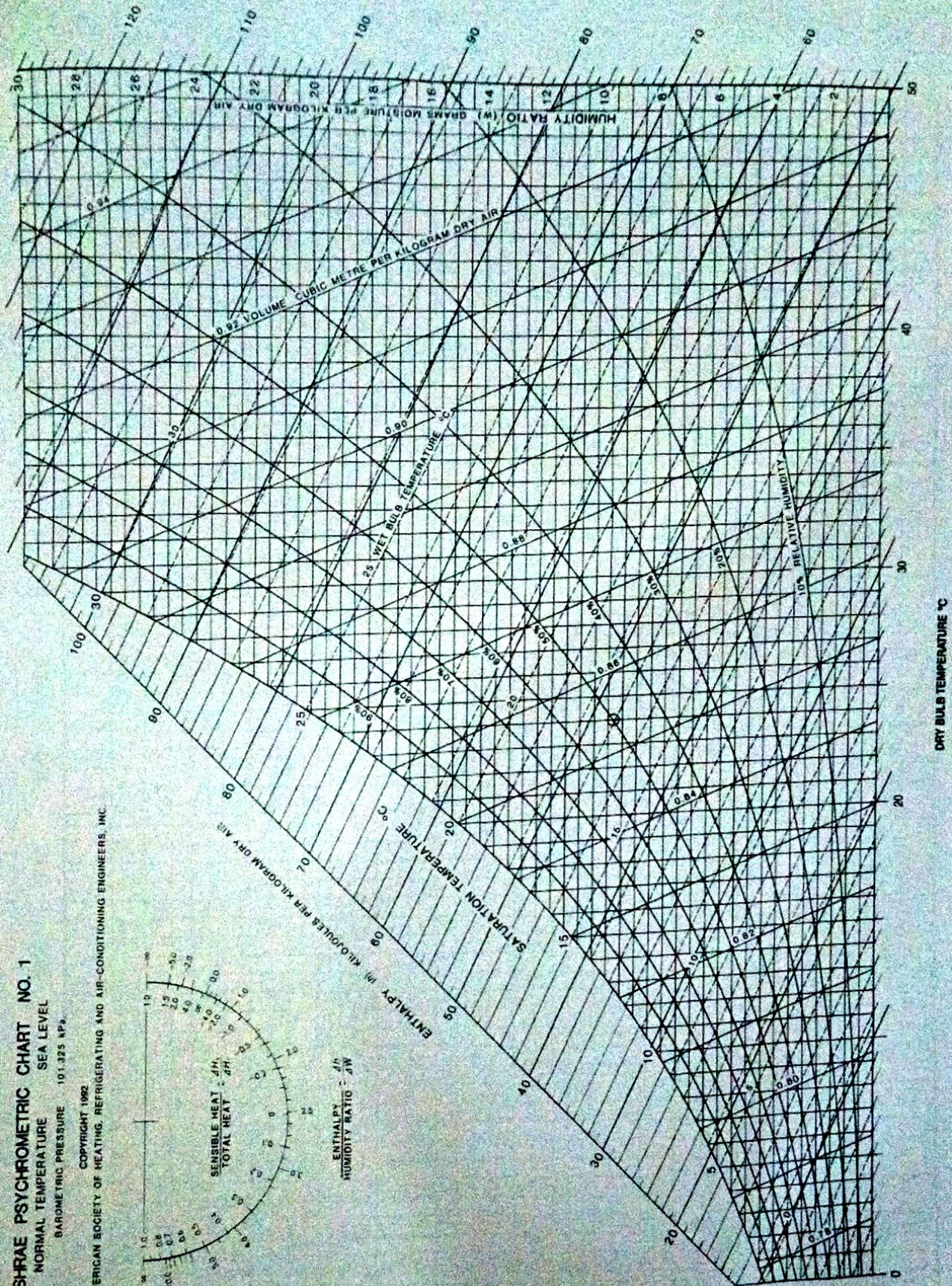
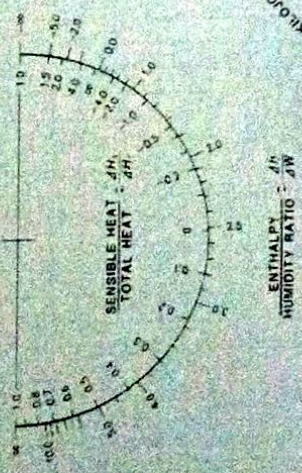
ASHRAE PSYCHROMETRIC CHART NO. 1

NORMAL TEMPERATURE SEA LEVEL

BAROMETRIC PRESSURE 101.325 kPa

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AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS, INC.



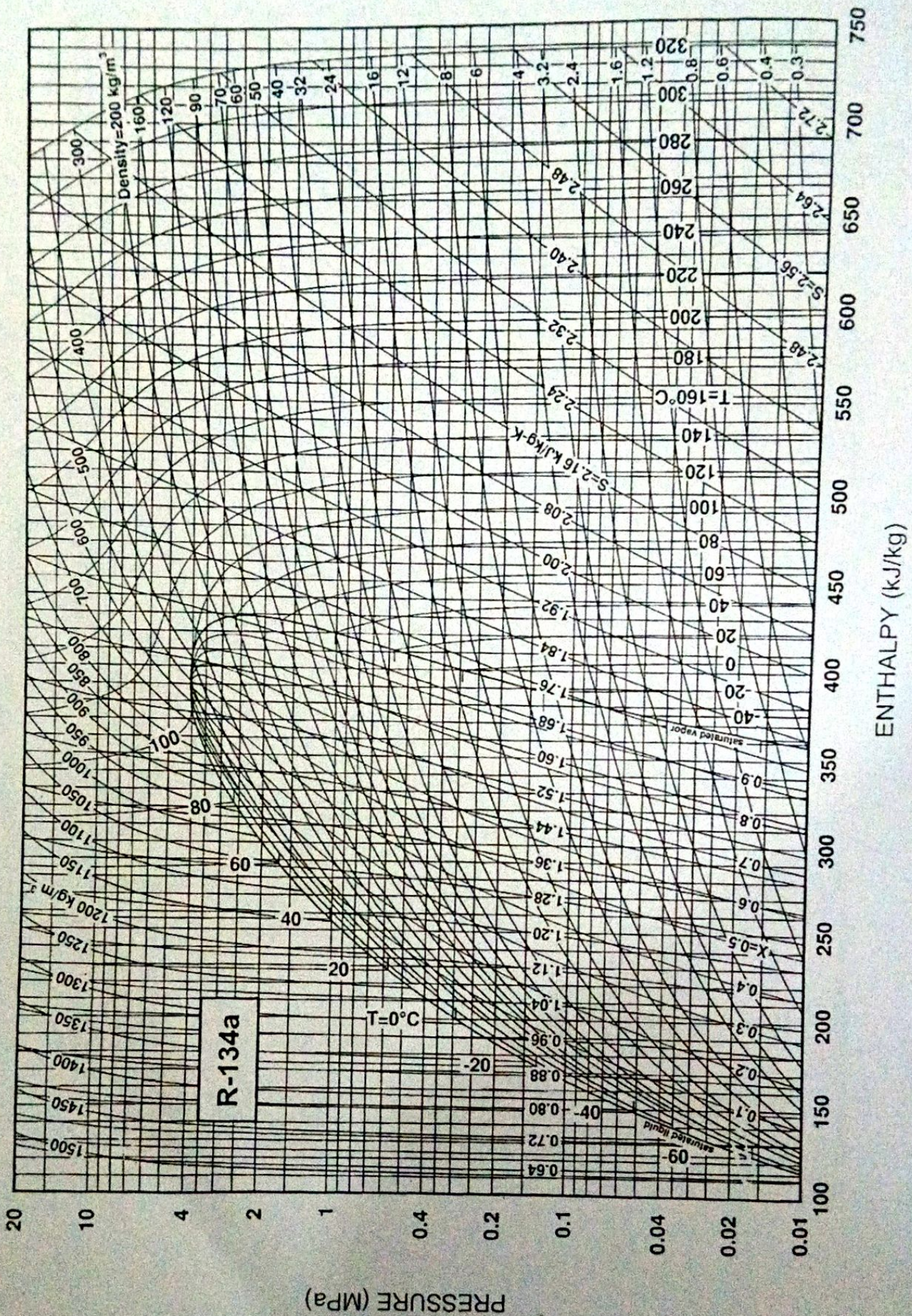


Fig. 8 Pressure-Enthalpy Diagram for Refrigerant 134a

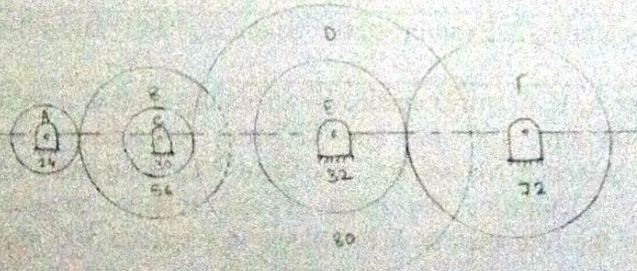


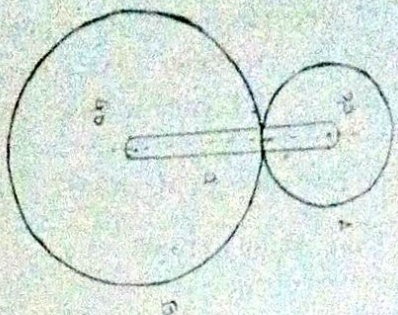
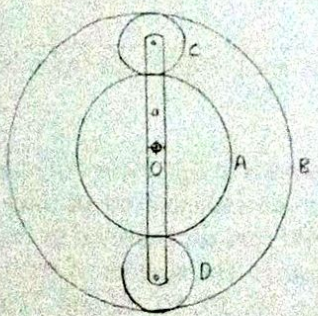
Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
Summer Term Examination
Academic Year: 2024-25

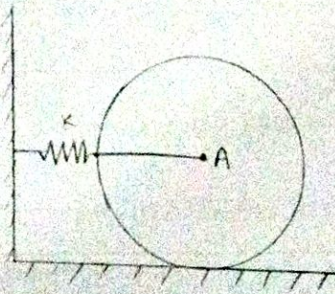
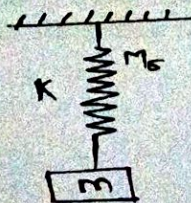
Class: TY B.Tech.	Subject: Dynamics of Machines	Subject Code: PCC-ME302
Date: 18.06.2025	Time: 10:00 to 13:00	Max. Marks: 100

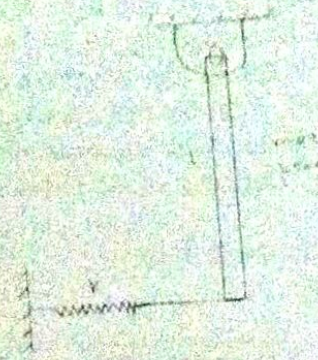
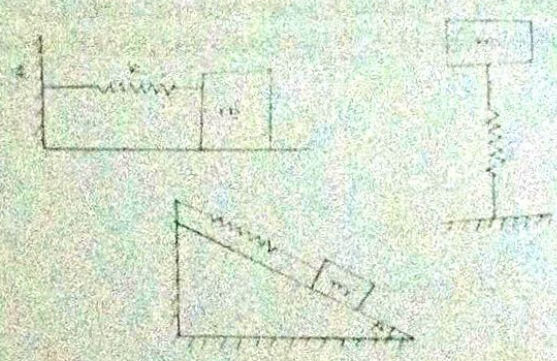
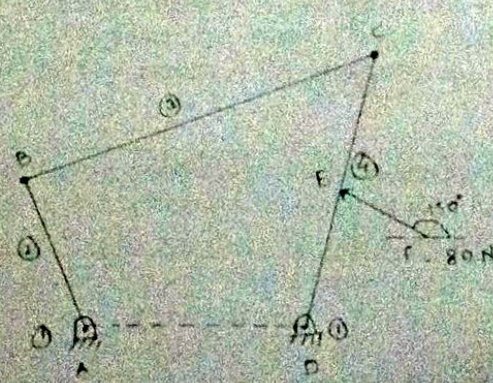
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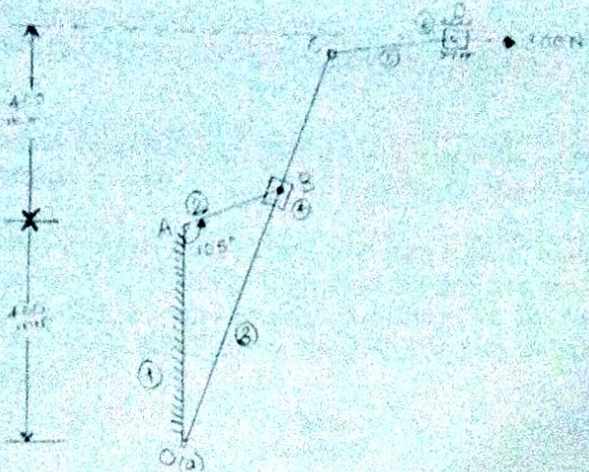
- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/sharing of stationery, calculator, etc. is not allowed.
- iv. Figures to the right indicate full marks.
- v. The symbols carry the usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	What is gear train? What are the different types of gear train? Explain it.	05	CO1, CO2	BT1, BT2
	b	A compound gear train shown in fig. consist of compound gear B-C and D-E. All gears are mounted on parallel shafts. The motor shaft rotating at 800 rpm is connected to the gear A and the output shaft to the gear F. The number of teeth on gears A, B, C, D, E and F are 24, 56, 30, 80, 32 and 72 respectively. Determine the speed of the gear F. 	03	CO1	BT3, BT4
	c	An epicyclic gear train consist of an arm and two gears A and B having 30 and 40 teeth respectively. The arm rotates about the center of the gear A at a speed of 80 rpm counterclockwise. Determine the speed of the gear B if (i) the gear A is fixed, and (ii) the gear A revolves at 240 rpm clockwise instead of being fixed.	05	CO1	BT3, BT4

					
	d	An epicyclic gear train is shown in fig. The number of teeth on A and B are 80 and 200. Determine the speed of the arm 'a' (i) if A rotates at 100 rpm clockwise and B at 50 rpm counterclockwise. (ii) if A rotates at 100 rpm clockwise and B is stationary. 	07	CO1	BT3, BT4
Q.2	a	What is balancing? Explain static balancing and Dynamic balancing.	06	CO4	BT1, BT2
	b	Four masses A, B, C and D carried by a rotating shaft at a radii 80 mm, 100 mm, 160 mm and 120 mm respectively are completely balanced. Masses B, C and D are 8 kg, 4 kg and 3 kg respectively. Determine the mass A and relative angular positions of the four masses if the planes are spaced 500 mm apart.	08	CO4	BT2, BT3
Q.3	a	Derive velocity and acceleration of piston for slider crank mechanism.	06	CO3	BT1, BT2
	b	The crank and connecting rod of a vertical petrol engine, running at 1800 rpm are 60 mm and 270 mm respectively. The diameter of the piston is 100 mm and the mass of the reciprocating parts is 1.2 Kg. During the expansion stroke when the crank has turned 20° from the top dead Centre the gas pressure is 650 kN/m². Determine the 1) Net force on the piston 2) Net load on the gudgeon pin 3) Thrust on the cylinder walls	07	CO3	BT2, BT3

	c	A horizontal gas engine running at 210 rpm has a bore of 220 mm and stroke of 440 mm. The connecting rod is 924 mm long and the reciprocating parts weigh 20 Kg. When the crank has turned through an angle of 30° from the inner dead centre, the gas pressure on the cover and the crank sides are 500 KN/m² and 60 KN/m² respectively. Diameter of the piston rod is 40 mm. Determine 1) Turning moment on the crank shaft 2) Trust on the bearings	07	CO3	BT2, BT3
Q.4	a	A uniform thin disk of mass 1 kg and radius 0.1 m is kept on a surface as shown in the figure. A spring of stiffness $k = 400$ N/m is connected to the disk centre A. Initially, the springs are unstretched. Assume pure rolling of the disk. For small disturbance from the equilibrium, the natural frequency of vibration of the system is _____ rad/s (round off to one decimal place) 	05	CO5	BT2, BT3
	b	Derive the natural frequency of spring mass system with spring having mass M_s for given system. 	08	CO5	BT3
	c	Find the natural frequencies of given vibrations system as rod having mass m and total length L. Spring stiffness K as shown in fig. (initially rod is vertical).	07	CO5	BT2, BT3

					
	d	Find the individual natural frequencies of given three vibrations system.	06	CO5	BT2, BT3
					
Q.5	a	A four-link mechanism with the following dimension is acted upon by a force $80\angle 150^\circ$ N on the link DC shown in fig. AD = 500 mm, AB = 400 mm, BC = 1000 mm, DC = 750 mm, DE = 350 mm. Determine the input torque T on the link AB for the static equilibrium of the mechanism for the given configuration. (Note: Angle made by link AB with horizontal link AD is 120°) 	06	CO3	BT2, BT3
	b	Write down the static equilibrium conditions for 1) Two force members 2) Three force members 3) Two force member and Torque	06	CO3	BT1

	c For the static equilibrium of the quick-return mechanism shown in fig. Determine the input torque T_2 to be applied on the link AB for a force of 300 N on the slider D. The dimensions of the various links are $OA = 400$ mm, $AB = 200$ mm, $OC = 800$ mm, $CD = 300$ mm. 	08	CO3	BT2, BT3
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=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Ability to analyze the gear trains used in various machines.
CO 2	Student demonstrates and analyzes different mechanisms used in various machines.
CO 3	Student exhibit skills towards application of static force analysis and synthesis of mechanisms.
CO 4	Student exhibits need, design and applications of balancing machine parts.
CO 5	Student exhibits solutions to issues related to vibration, their effect and its control measures.

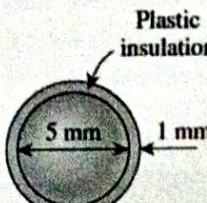


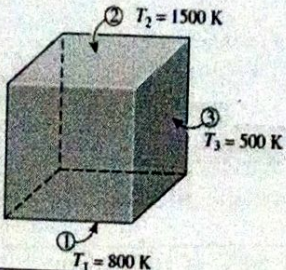
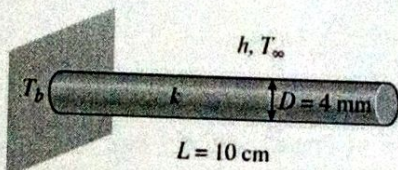
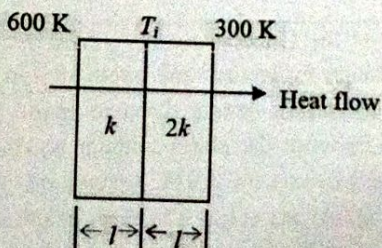
Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
Summer Term Examination
Academic Year: 2024-25 (Semester -I)

Class: TY	Subject: Heat Transfer	Subject Code: PCC-ME-301
Date: 17.06.2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- i. All questions are compulsory.
- ii. Assume suitable data whenever necessary and state it clearly.
- iii. Use of non-programmable calculator is permitted. This paper contains two pages.

Q.N.		Marks	CO	BT
Q.1	Explain, using clear reasoning or an equation, how temperature affects the thermal conductivity of silver (a metal), asbestos (a non-metal), and hydrogen gas.	7	CO1	BT 1-3
Q.2	A 5-mm-diameter spherical ball at 50°C is covered by a 1-mm-thick plastic insulation ($k=0.13 \text{ W/m}\cdot\text{K}$). The ball is exposed to a medium at 15°C, with a convection heat transfer coefficient of $20 \text{ W/m}^2\cdot\text{K}$. Determine if the plastic insulation on the ball will help or hurt heat transfer from the ball. 	8	CO1	BT 3-4
Q.3	What is the physical significance of Prandtl number? Plot the thermal boundary layer and hydrodynamic boundary layer for the case of $Pr < 1$.	7	CO2	BT 1-2
Q.4	A solid sphere of radius 10 mm is placed at the centroid of a hollow cubical enclosure of side length 30 mm. The outer surface of the sphere is denoted by 1 and the inner surface of the cube is denoted by 2. The view factor F_{22} for radiation heat transfer is _____?	8	CO2	BT 3-4
Q.5	What is the physical significance of the Grashof number? What is the Rayleigh number, and what are its ranges for laminar and turbulent flow?	7	CO2	BT 1-2
Q.6	Two infinite parallel plates are placed at a certain distance apart. An infinite radiation shield is inserted between the plates without touching any of them to reduce heat exchange between the plates. Assume that the emissivities of plates and radiation shield are equal. The ratio of the net heat exchange between the plates with and without the shield is _____?	8	CO2	BT 3-4
Q.7	A stream of hydrocarbon is cooled from 160° C to 60° C in the tube of double pipe counter flow heat exchange having mass flow rate of 700 kg/h. Water with a mass flow rate of 500 kg/h enters the heat exchange at 10° C. The outer diameter of the inner tube is 2.5 cm, and the length of the tube is 6 m. The specific heat capacities of the hydrocarbon and water are 2.2 kJ/kg·K and 4.18 kJ/kg·K, respectively. Calculate the following: a) The outlet temperature of the water. b) The logarithmic mean temperature difference (LMTD). The overall heat transfer coefficient based on the outer surface.	15	CO 3-4	BT 3-4

Q.8	Consider the 5-m X 5-m X 5-m cubical furnace shown in Figure, whose surfaces closely approximate black surfaces. The base, top, and side surfaces of the furnace are maintained at uniform temperatures of 800 K, 1500 K, and 500 K, respectively. Determine (a) the net rate of radiation heat transfer between the base and the side surfaces, (b) the net rate of radiation heat transfer between the base and the top surface, and (c) the net radiation heat transfer from the base surface. Note: the value of $F_{1-2} = 0.2$. 	15	CO 2-3	BT 3-4
Q.9	c) A 4-mm-diameter and 10-cm-long aluminum fin ($k=237 \text{ W/m}\cdot\text{K}$) is attached to a surface. If the heat transfer coefficient is $12 \text{ W/m}^2\cdot\text{K}$, determine the percent error in the rate of heat transfer from the fin when the infinitely long fin assumption is used instead of the adiabatic fin tip assumption. 	15	CO 1-3	BT 3-4
Q.10	Heat transfer through a composite wall is shown in figure. Both the sections of the wall have equal thickness (l). The conductivity of one section is k and that of the other is $2k$. The left face of the wall is at 600 K and the right face is at 300 K.  Find out the interface temperature T_i (in K) of the composite wall? == ALL THE BEST ==	10	CO 1-2	BT 3-4

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Student will be able to analyze heat conduction with and without internal heat generation, critical thickness of insulation and extended surfaces with the practical utilities.
CO 2	Student demonstrates the concept and mechanism of convection, conduction and radiation.
CO 3	Student exhibits application of mathematical approach to analyze and solve the numerical on complex heat transfer phenomena's.
CO 4	Student can explain the design, performance analysis and practical applications of heat exchangers.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Summer Term (Semester-II) 2024-25

S.Y. B. Tech.

Mechanical Engineering Department

Fluid Mechanics and Fluid Machines / FMHM

PCC-ME202 / PCC-ME207

Date: 25/06/25

Time: 14:30 to 17:30 Hrs

Max. Marks :100

Note: i. Solve All Questions ii. Assume suitable data if necessary. iii. Use of a non-programmable calculator is permitted.

Q.N.	Question	Marks	CO	BT
Q.1	(A) The velocity distribution for flow over a flat plate is given $u = \frac{3}{4}y - y^2$ in which u is the velocity in m/s at a distance y meter above the plate. Calculate Shear stress at $y = 0.1$ m (Viscosity for fluid = 8.5 poise)	8	1	4
	(B) Define the terms: Density, specific volume, specific gravity, Surface tension, vacuum pressure, compressible and incompressible fluid	7	1	1
Q.2	(A) The diameters of a pipe at sections 1 and section 2 are 15 cm and 20 cm respectively. Find the discharge flow through the pipe if the velocity of water at section 1 is 4 m/s. Also determine the velocity at section 2.	5	2	3
	(B) Enlist and define any 10 types of fluid flow.	5	2	3
	(C) What is Bernoulli's theorem? State the assumptions made in Bernoulli's Equation	5	3	2
Q.3	(A) Define an Venturimeter meter. Derive an equation for discharge using Bernoulli's Equation.	8	3	2
	(B) Obtain an expression for force exerted by the jet of water on fixed flat plate in the direction of the Jet.	7	4	3
Q.4	(A) The rate of flow of water through horizontal pipe is $0.3 \text{ m}^3/\text{s}$. The diameter of pipe is suddenly enlarged from 0.25 m to 0.5 m. Calculate head load due to sudden enlargement and power lost due to enlargement.	8	4	3
	(B) A jet of water striking on an unsymmetrical moving curved plate when the jet strikes tangentially at one of the tips. Draw the velocity triangle considering flow angle at outlet greater than 90° . Give the nomenclature of all the velocities and angles.	7	5	2
Q.5	(A) Show that the efficiency of a free jet striking normally on a series of flat plates mounted on the periphery of a wheel never exceeds 50%.	10	4	3
Q.6	(A) The external and internal diameters of an inward flow reaction turbine are 1.20 m and 0.6 m respectively. The head on the turbine is 22 m and velocity of flow through the runner is constant and equal to 2.5 m/s. The guide blade angle is given as 10° and the runner vanes are radial at inlet. If the discharge at outlet is radial, determine: (i) The speed of the turbine, (ii) The vane angle at outlet of the runner, and (iii) Hydraulic efficiency.	10	5	4
	(B) What is draft tube? What are its functions?	5	5	2
Q.7	(A) The internal and external diameters of the impeller of a centrifugal pump are 200 mm and 400 mm respectively. The pump is running at 1200 r.p.m. The vane angles of the impeller at inlet and outlet are 20° and 30° respectively. The water enters the impeller radially and velocity of flow is constant. Determine the work done by the impeller per unit weight of water.	10	5	4
	(B) What are different casings used in centrifugal pumps? Explain any one of them.	5	5	2

Course Outcomes (CO)

CO1. Apply properties of fluid to predict the behaviour of the fluids in various engineering applications. CO2. Apply principles of conservation of mass, momentum, and energy to analyze flow behaviour. CO3. Apply Bernoulli's equation to different applications. CO4. Demonstrate basic knowledge about turbines and pumps through the impact of the jet. CO5. Evaluate work done and efficiency of different turbines and pumps.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create